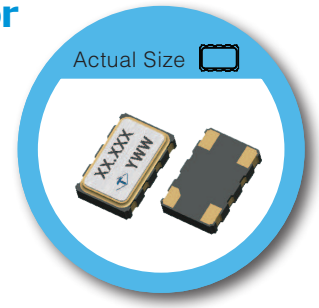
TV Type
5.0 x 3.2 mm SMD Voltage Controlled Temperature Compensated Crystal Oscillator



FEATURE

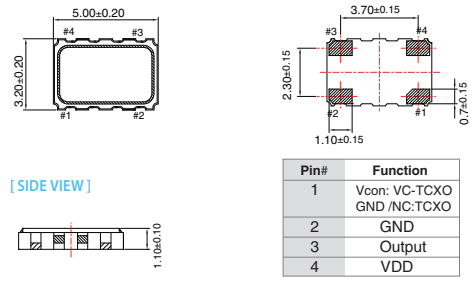
- Typical 5.0 x 3.2 x 1.1 mm ceramic SMD package.
- For automatic assembly
- Compactness and light weight
- Low Power consumption.
- Clipped Sine wave (without DC-Cut capacitor) output optional.
- Voltage control TCXO available
- -40 to 85°C operation

TYPICAL APPLICATION

- WLAN - Telecommunication - Mobile Application

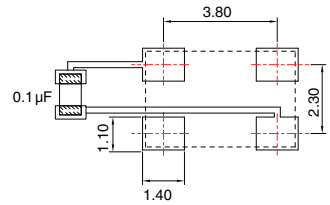
RoHS Compliant

DIMENSION (mm)



Pin#	Function
1	Vcon: VC-TCXO GND /NC:TCXO
2	GND
3	Output
4	VDD

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance,
place a by-pass capacitor of 0.1µF
as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter		3.3V/3.0V/2.8V		2.5V		1.8V		Unit
		Min.	Max.	Min.	Max.	Min.	Max.	
Supply Voltage Variation (V_{DD})		2.66	3.465	2.375	2.625	1.71	1.89	V
Frequency Range		10	52	10	52	10	52	MHz
Standard Frequency		10, 12.8, 13, 16.368, 16.369, 19.2, 19.44, 20, 25, 26, 27, 30, 30.72, 32, 38.4						MHz
Frequency Tolerance		-	±2.0	-	±2.0	-	±2.0	ppm
Frequency Stability								
vs Supply Voltage (±5%) change		-	±0.2	-	±0.2	-	±0.2	ppm
vs Load (±10%) change		-	±0.2	-	±0.2	-	±0.2	ppm
vs Aging (@1st year)		-	±1.0	-	±1.0	-	±1.0	ppm
Supply Current (Clipped Sine)								
10MHz ≦ Fo ≦ 26MHz		-	1.5	-	1.5	-	1.5	mA
26MHz < Fo ≦ 52MHz		-	2	-	2	-	2	mA
Clipped Sine	Output Level	0.8	-	0.8	-	0.8	-	V
	Load	10KΩ//10pF		10KΩ//10pF		10KΩ//10pF		
Control Voltage Range		0.5	2.5	0.4	2.4	0.3	1.5	V
Pulling Range		±5	-	±5	-	±5	-	ppm
Vc Input Impedance		500	-	500	-	500	-	kΩ
Phase Noise @Fo=19.2M Clipped Sine	@100Hz	-115		-115		-115		dBc/Hz
	@1KHz	-135		-135		-135		dBc/Hz
	@10KHz	-148		-148		-148		dBc/Hz
Start Time		-	2	-	2	-	2	mSec
Storage Temp. Range		-40	85	-40	85	-40	85	°C

Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position.
*Frequency at 25°C, 1 hour after reflow.

FREQ.STABILITY vs. TEMP. RANGE

Temp. (°C) \ ppm	±0.5	±1.0	±1.5	±2.0	±2.5
-20 ~ +70	○	○	○	○	○
-30 ~ +85	○	○	○	○	○
-40 ~ +85	○	○	○	○	○

* O:Available △:Conditional X: Not available
*Inclusive of calibration @ 25°C, operating tempera ture range, input voltage variation,load variation, aging (1st year), shock, and vibration

Note: not all combination of options are available. Other specifications may be available upon request.